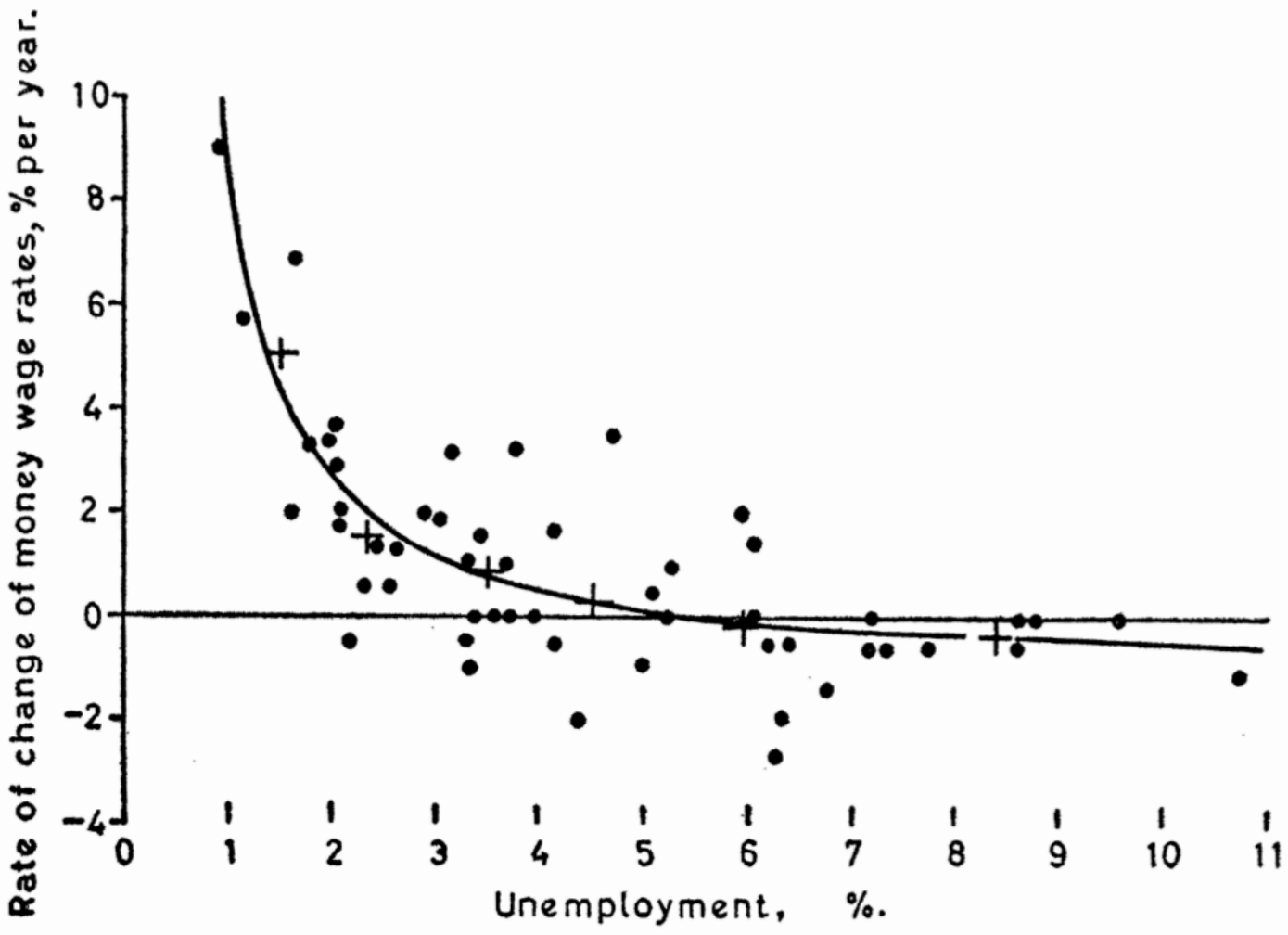
# BEVERIDGEAN PHILLIPS CURVE

Pascal Michaillat

November 2026

Available at https://pascalmichaillat.org/15/

# TRADITIONAL PHILLIPS CURVE LINKED INFLATION TO UNEMPLOYMENT (UK, 1861–1913)

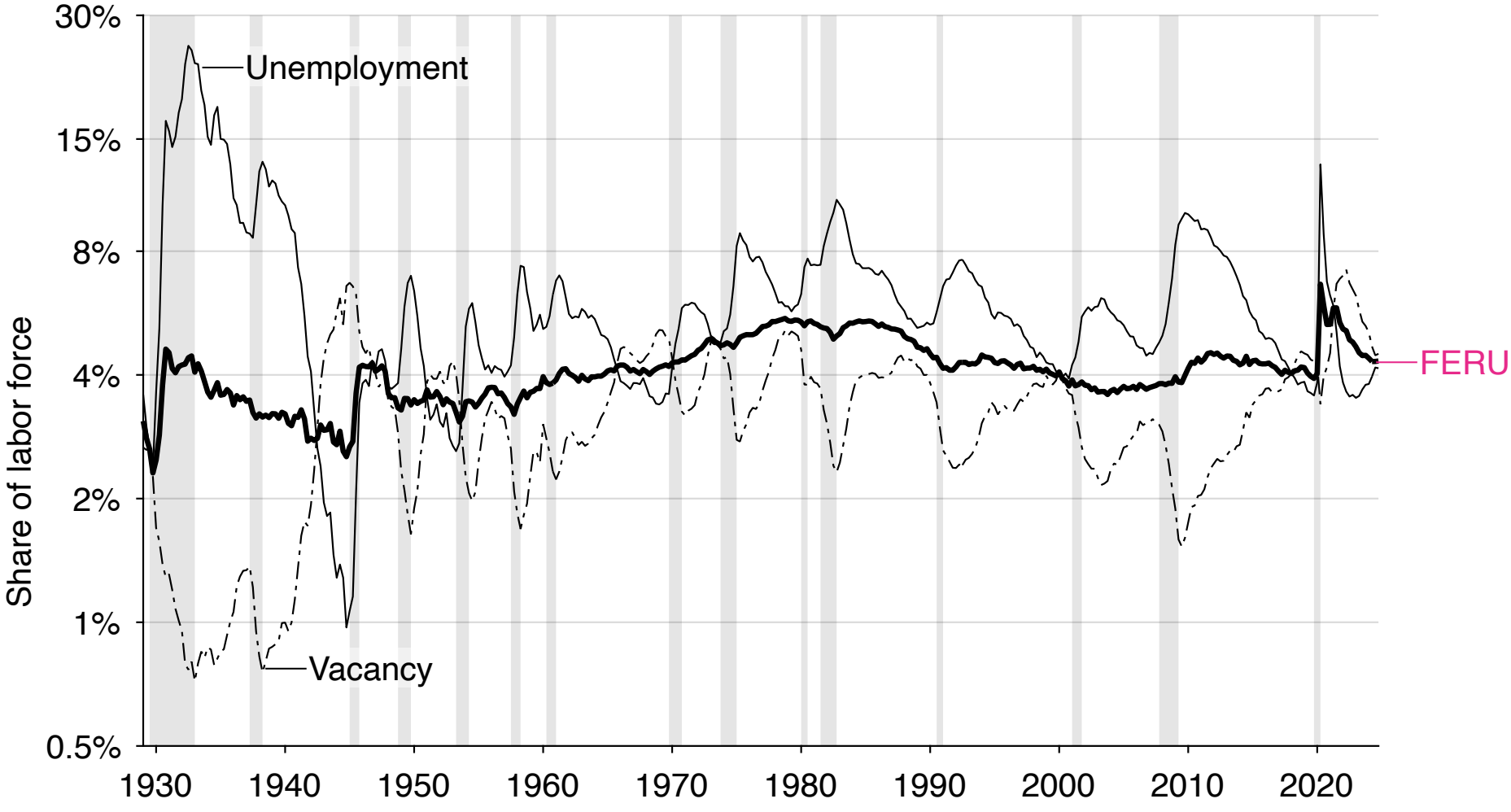


# UNEMPLOYMENT VANISHED FROM MODERN PHILLIPS CURVE

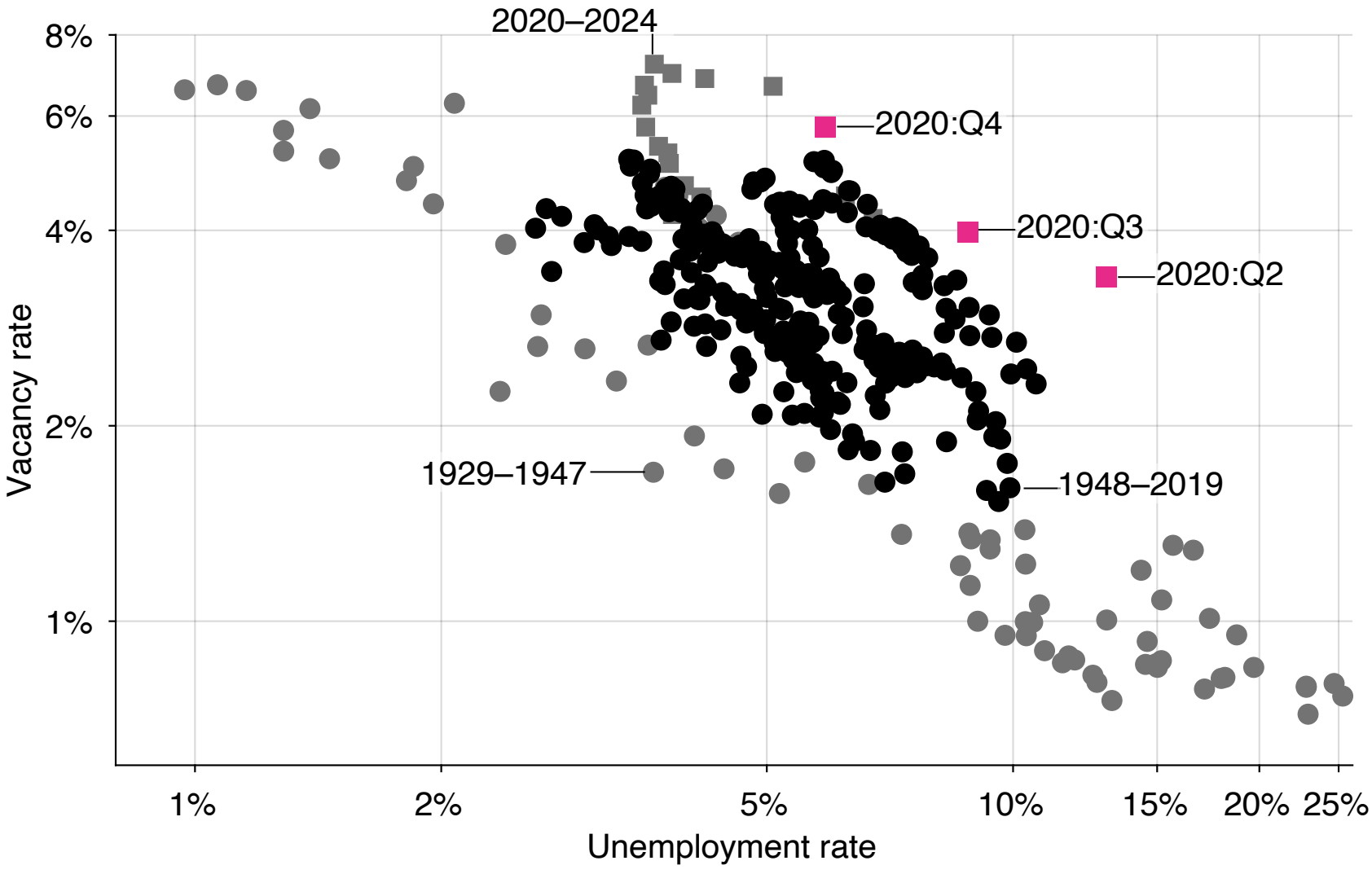
- New Keynesian Phillips curve:
  - Output/employment $\rightsquigarrow$ marginal costs $\rightsquigarrow$ inflation
- Blind spots for monetary policy:
  - Are full employment and price stability consistent?
  - What is the tradeoff between inflation and unemployment?
  - Do labor market disruptions impact inflation?
  - Why doesn't inflation fall more in recessions?
  - How much might inflation spike in booms?
- This paper: model of the Phillips curve with unemployment

# CASE STUDY: POST-PANDEMIC MONETARY POLICY

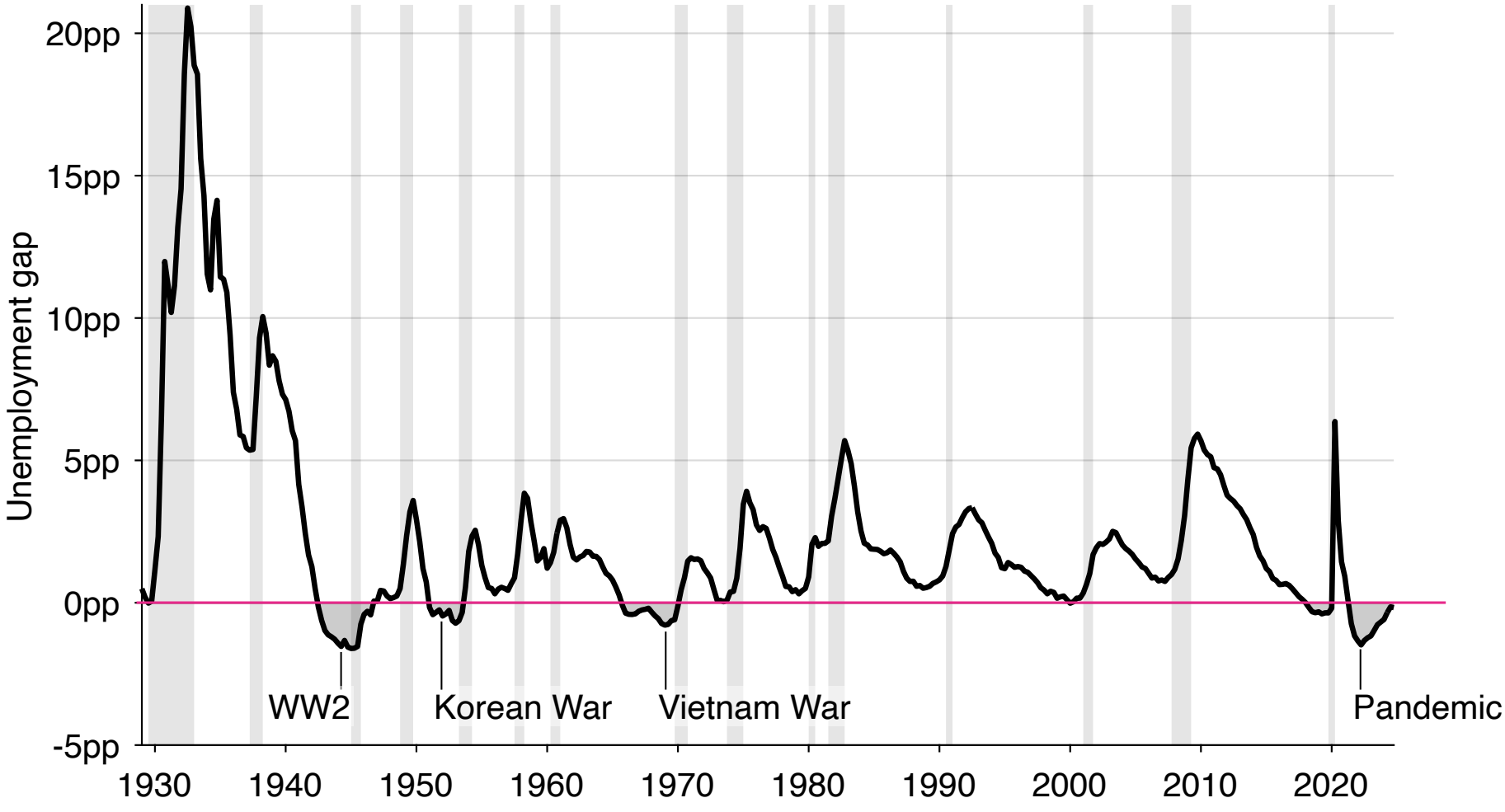
# FERU IS USUALLY STABLE BUT SPIKED IN 2020 ...



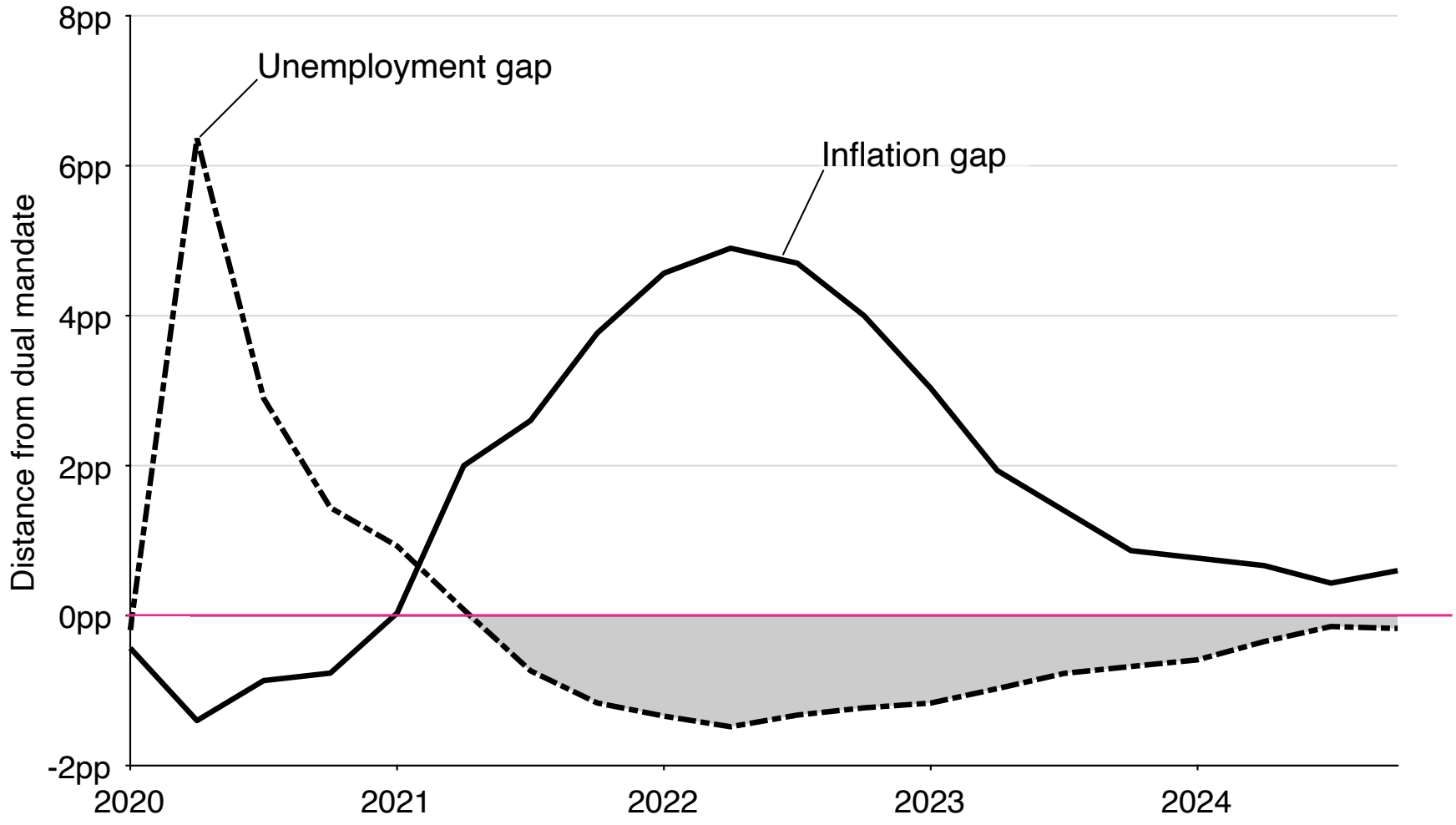
# . . . BECAUSE BEVERIDGE CURVE SHIFTED OUTWARD



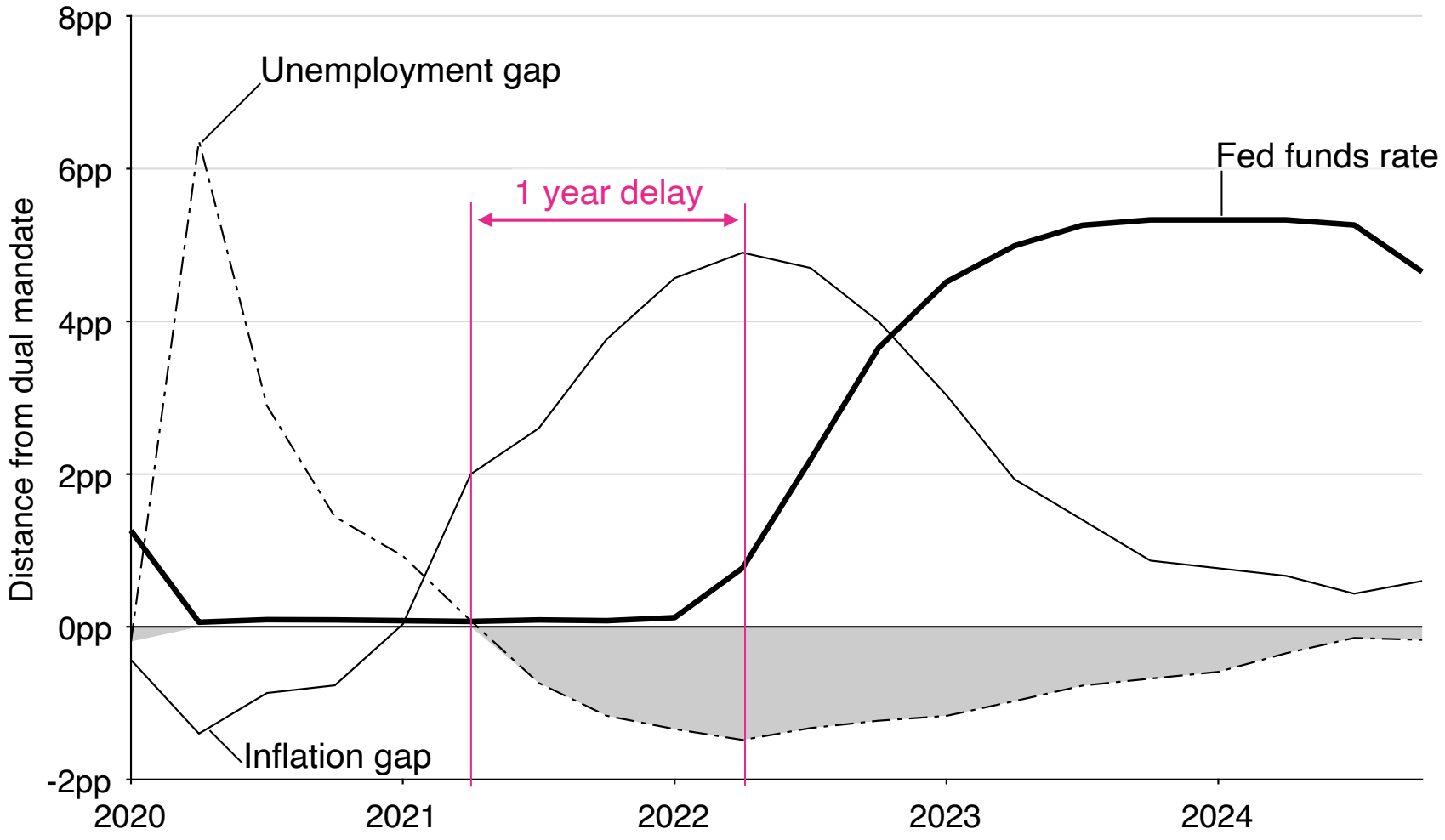
# . . . CREATING RARE ABOVE-FULL-EMPLOYMENT ECONOMY



# . . . AND GENERATING SUSTAINED BURST OF INFLATION



# . . . WHICH THE FED MIGHT NOT HAVE UNDERSTOOD

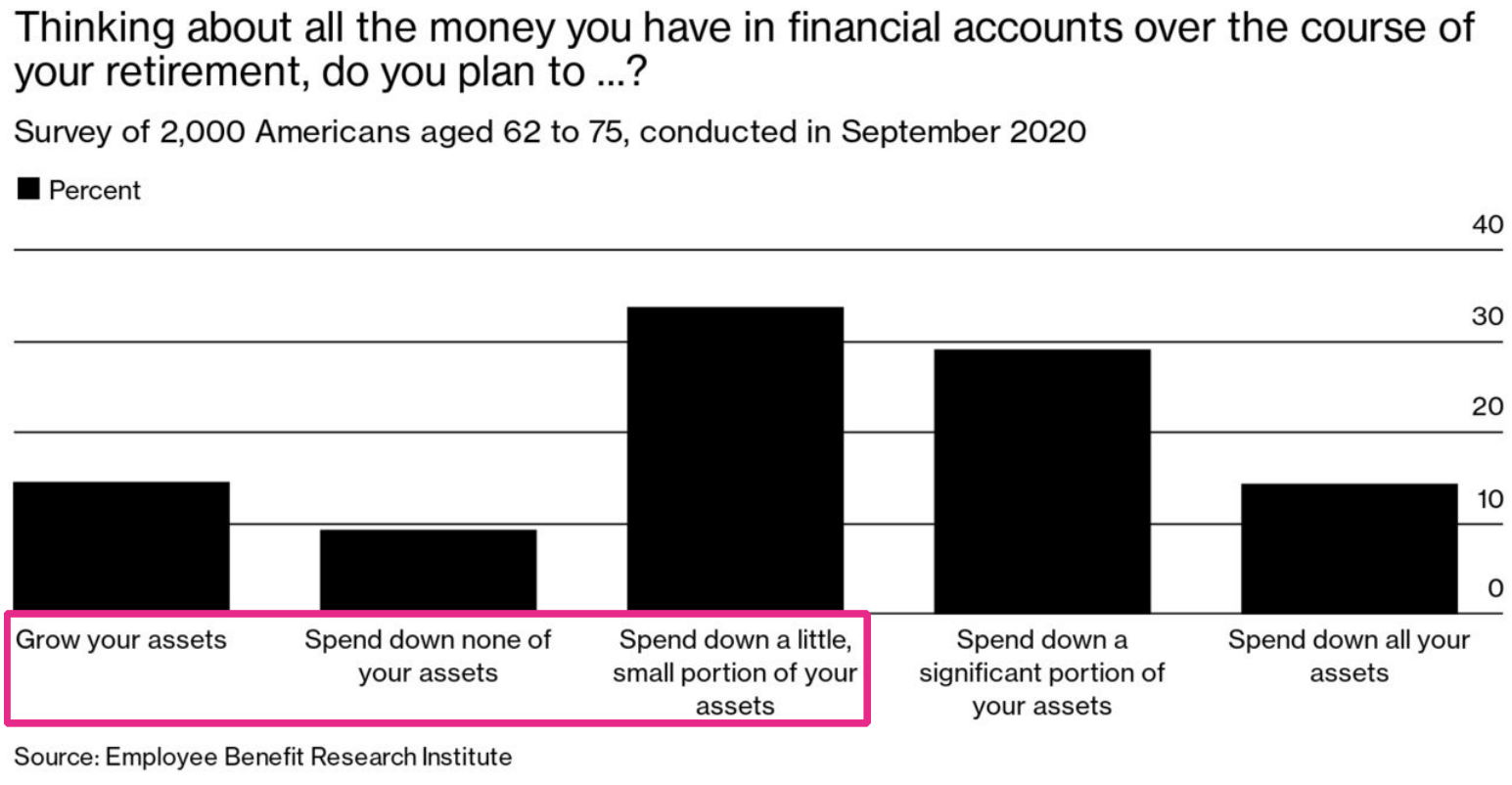


# DESCRIPTION OF THE MODEL

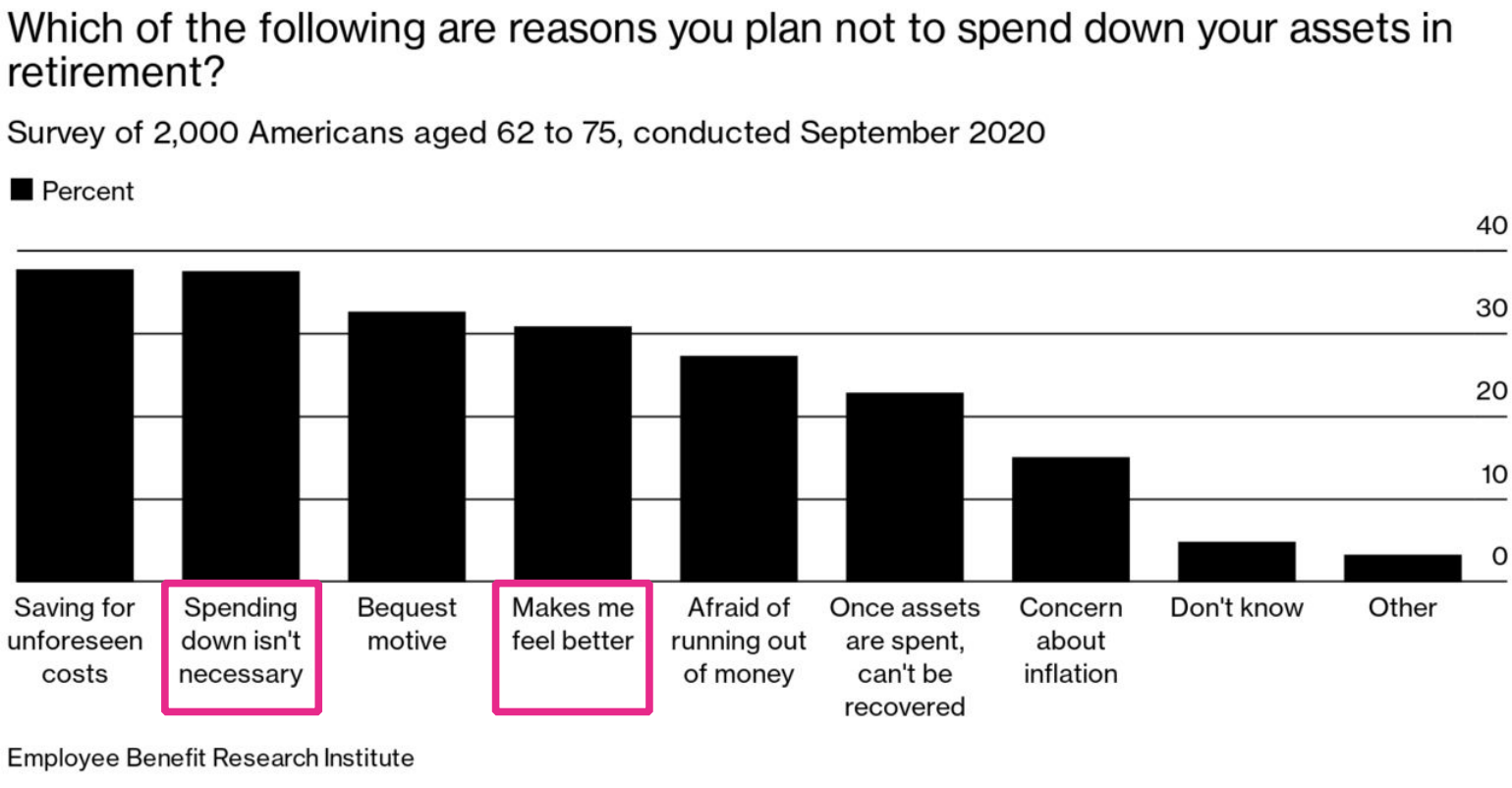
# OVERVIEW OF THE ASSUMPTIONS

1. Households buy services on matching market $\rightsquigarrow$ unemployment
   - Michaillat & Saez (2022)
2. Households derive utility from wealth $\rightsquigarrow$ nondegenerate AD curve
   - Michaillat & Saez (2021)
3. Job search is directed $\rightsquigarrow$ inflation-unemployment tradeoff
   - Moen (1997)
4. Quadratic price-adjustment costs $\rightsquigarrow$ nonvertical Phillips curve
   - Rotemberg (1982)

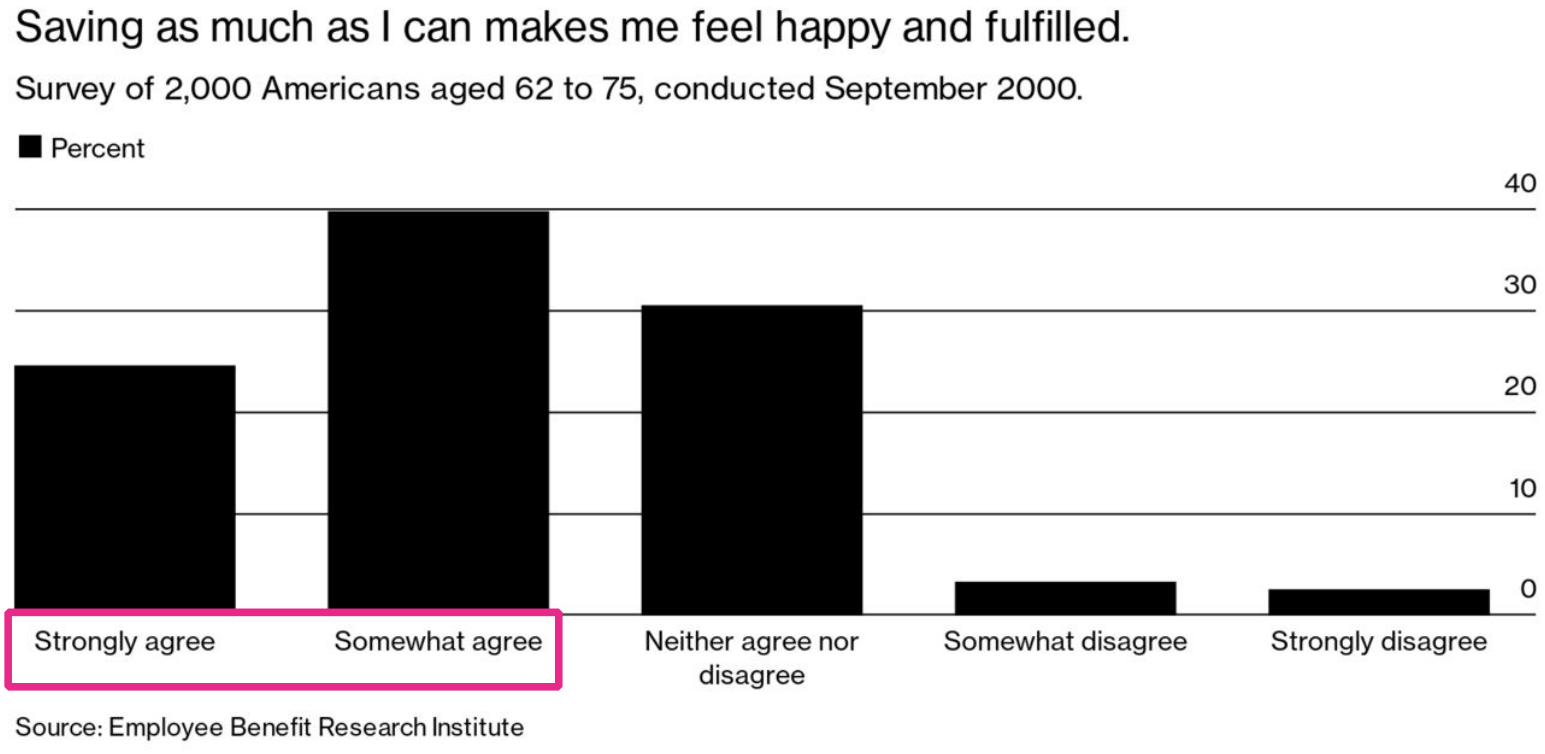
“THE VIRTUE OF THE CAKE WAS THAT IT WAS NEVER TO BE CONSUMED, NEITHER BY YOU NOR BY YOUR CHILDREN AFTER YOU” (KEYNES 1919)



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# HIGHER-THAN-NORMAL PRICE GROWTH UPSETS CUSTOMERS (SHILLER 1997)

A7. When you go to the store and see that prices are higher, do you sometimes feel a little angry at someone?

[Circle one number]

| | | |
|---|---|---|
| 1. Yes, often | [38%] | [$n$=120] |
| 2. Yes, sometimes | [48%] | |
| 3. Never | [15%] | |

# HIGHER-THAN-NORMAL PRICE GROWTH UPSETS CUSTOMERS (SHILLER 1997)

While people appear to be in great disagreement why inflation occurs, they do tend nonetheless to be angry at someone when they see prices rise: in answer to question A7, 38% reported feeling angry often, and an additional 48% reported feeling angry sometimes. There was little agreement on the answer to question A8, on who they are angry at. The government was mentioned by 18 respondents, manufacturers by 15, store owners by 6, business in general by 6, wholesalers by 4, executives by 3, the U.S. Congress by 3, and greedy people by 3. Also mentioned were institutions, economists, retailers, distributors, middlemen, conglomerates, the president of the United States, the Democratic Party, big-money people, store employees (for wage demands), "my employer" (for not increasing my salary), and myself (for being ignorant of matters).

# HIGHER-THAN-NORMAL PRICE GROWTH UPSETS CUSTOMERS (SHILLER 1997)

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# LOWER-THAN-NORMAL PAY GROWTH DAMAGES MORALE (BEWLEY 1999)

TABLE 11.2 Arguments against cutting base pay (applies to 151 businesses)

| Arguments | Number of businesses | Percentage of businesses |
|---|---|---|
| Pay cuts hurt morale and demotivate workers | 104 | 69 |
| Because of lower living standards | 38 | 25 |
| Because they are insulting | 26 | 17 |
| Pay cuts hurt productivity | 63 | 42 |
| Pay cuts increase turnover | 62 | 41 |
| Employees would leave slowly or at the onset of the next boom | 45 | 30 |
| The best employees would leave | 18 | 12 |
| Employees would quit out of rage | 5 | 3 |
| It is hard to convince workers that a pay cut is needed to save their jobs | 11 | 7 |
| Pay cuts invite sabotage | 8 | 5 |
| Pay cuts invite unionization | 7 | 5 |
| Unions resist pay cuts | 4 | 3 |
| Managers do not want to cut their own pay | 5 | 3 |

# LOWER-THAN-NORMAL PAY GROWTH DAMAGES MORALE (BEWLEY 1999)

It's a personal thing. People can't explain a pay cut at home, whereas if they are laid off they can draw unemployment benefits and look for new work. A pay cut is like a criticism or an insult. Pay is so closely associated with self-worth that a cut is taken personally.

—Owner of a nonunion electronics manufacturing company with 15 employees

I have never cut anyone's pay. I don't believe in it in principle. . . . A pay cut would be interpreted as a punishment, even if it were done across the board. It would be insulting and would lower people's standard of living and for both those reasons, it would hurt morale and get people working against rather than for the restaurant. In this business, that could happen in a couple of days.

—Manager of a restaurant with 30 employees

I know something real. Never cut wages. If you do, you make enemies. They don't understand economics. They will wonder what they did wrong. And enemies hire lawyers.

—President of a nonunion manufacturing company with 40 employees

Morale suffers from a pay cut for psychic reasons. It says, "I am doing badly," even if everyone gets a pay cut. It is part of American culture. Individual performance is identified with how well you are paid. If your pay was cut, your neighbors would view you as working for a bad company.

—High-level human resource official in an insurance company with 50,000 employees

# SOLVING THE MODEL

# HOUSEHOLD'S PROBLEM

- Household $j$ maximizes utility

$$\int_0^{\infty} e^{-\delta t}\left[2a\sqrt{c_j(t)} + \frac{B_j(t) - B(t)}{P(t)} - \frac{\omega}{2}\left(\pi_j(t) - \pi^n\right)^2\right] dt$$

- Subject to law of motion of nominal wealth:

$$\dot{B}_j = iB_j + \int_0^1 P_i\left[1 - u(\theta_i)\right] kh_j\, di - P_j\left[1 + \tau(\theta_j(P_j))\right] c_j - T$$

- And law of motion of the paid service price:

$$\dot{P}_j(t) = \pi_j(t) \cdot P_j(t)$$

# SOLVING HOUSEHOLD'S PROBLEM BY HAMILTONIAN

- In real terms, Hamiltonian of household $j$'s maximization is

$$\begin{aligned}\mathcal{H} = {} & 2a\sqrt{c_j} + (b_j - b) - \frac{\varpi}{2}\left(\pi_j - \pi^n\right)^2 \\ & + \psi_j^b \left\{ r b_j + \left[1 - u(\theta)\right] k h_j - \left[1 + \tau(\theta_j(p_j))\right] p_j c_j - \frac{T}{P} \right\} \\ & + \psi_j^p \left(\pi_j - \pi\right) p_j\end{aligned}$$

- Control variables: $c_j, \pi_j$
- State variables: $b_j, p_j$
- Costate variables: $\psi_j^b, \psi_j^p$

## FIRST-ORDER CONDITIONS IN SYMMETRIC SOLUTION

- Market equilibrium $\rightsquigarrow$ AS curve

$$y = [1 - u(\theta)] \cdot kh$$

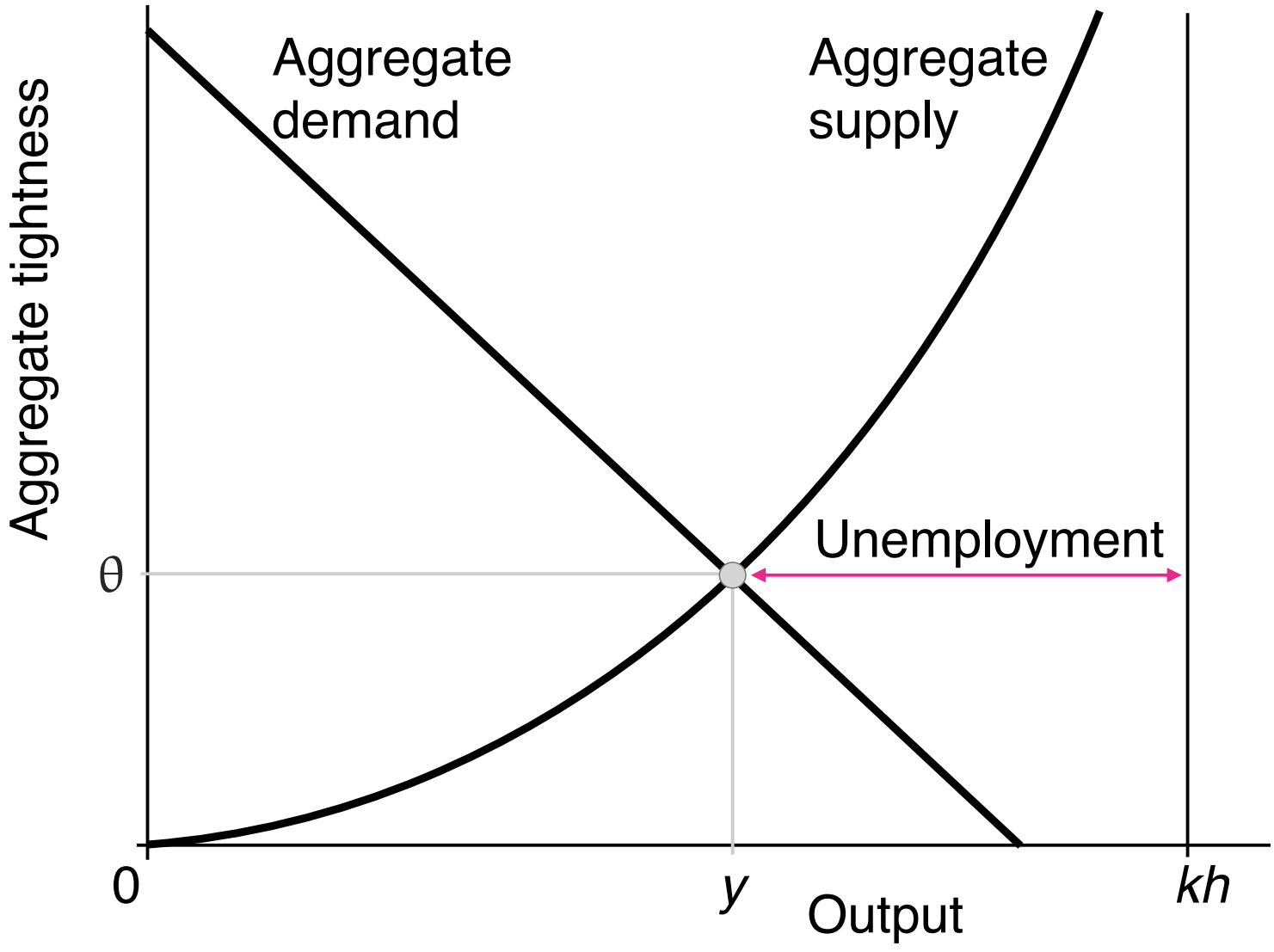
- Consumption-saving block $\rightsquigarrow$ AD curve

$$\frac{a}{\sqrt{c}} = \psi^b \cdot [1 + \tau(\theta)]$$
$$\dot{\psi}^b = (\delta - r) \cdot \psi^b - 1$$

- Service-pricing block $\rightsquigarrow$ Phillips curve

$$\varpi \cdot \left(\pi - \pi^n\right) = \psi^p$$
$$\dot{\psi}^p = \delta\psi^p + \psi^b y \cdot \left[1 - \frac{\eta(\theta)}{1 - \eta(\theta)} \cdot \frac{\tau(\theta)}{u(\theta)}\right]$$

# AGGREGATE DEMAND & AGGREGATE SUPPLY



# #1 - DIVINE COINCIDENCE

# PRICE STABILITY ⇔ FULL EMPLOYMENT

- Costate $\psi^p$ jumps to equilibrium point $\rightsquigarrow$ Phillips curve:

$$\pi = \pi^* + \frac{y(\theta)}{\varpi\delta(\delta - r)} \cdot \left[ \frac{\eta(\theta)}{1 - \eta(\theta)} \cdot \frac{\tau(\theta)}{u(\theta)} - 1 \right]$$

- Social efficiency minimizes labor devoted to recruiting & jobseeking:

$$\frac{\tau(\theta)}{u(\theta)} = \frac{1 - \eta(\theta)}{\eta(\theta)}$$

$\rightsquigarrow$ Inflation is on target ($\pi = \pi^*$) whenever tightness is efficient ($\theta = \theta^*$) and unemployment is efficient ($u = u(\theta^*) =$ FERU)

# DIVINE COINCIDENCE BREAKS WITH POOR MONETARY COMMUNICATION

- Ineffective communication fails to anchor normal inflation: $\pi^n \neq \pi^*$
- Policy error term appears in Phillips curve:

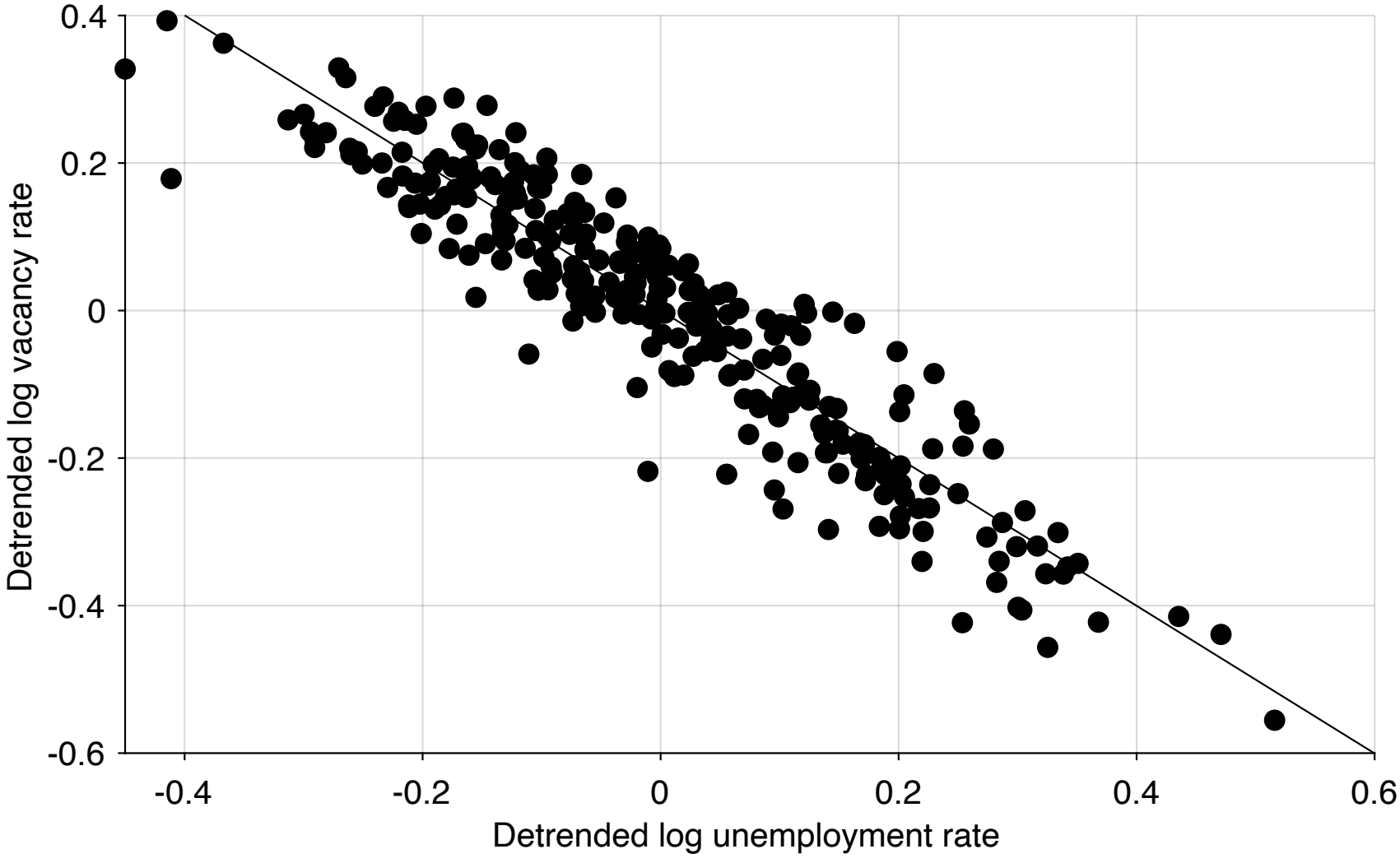
$$\pi = (\pi^n - \pi^*) + \pi^* + \frac{y(\theta)}{\varpi\delta(\delta - r)}\left[\frac{\eta(\theta)}{1-\eta(\theta)} \cdot \frac{\tau(\theta)}{u(\theta)} - 1\right]$$

- Since mid-1990s: US inflation expectations stable around 2%

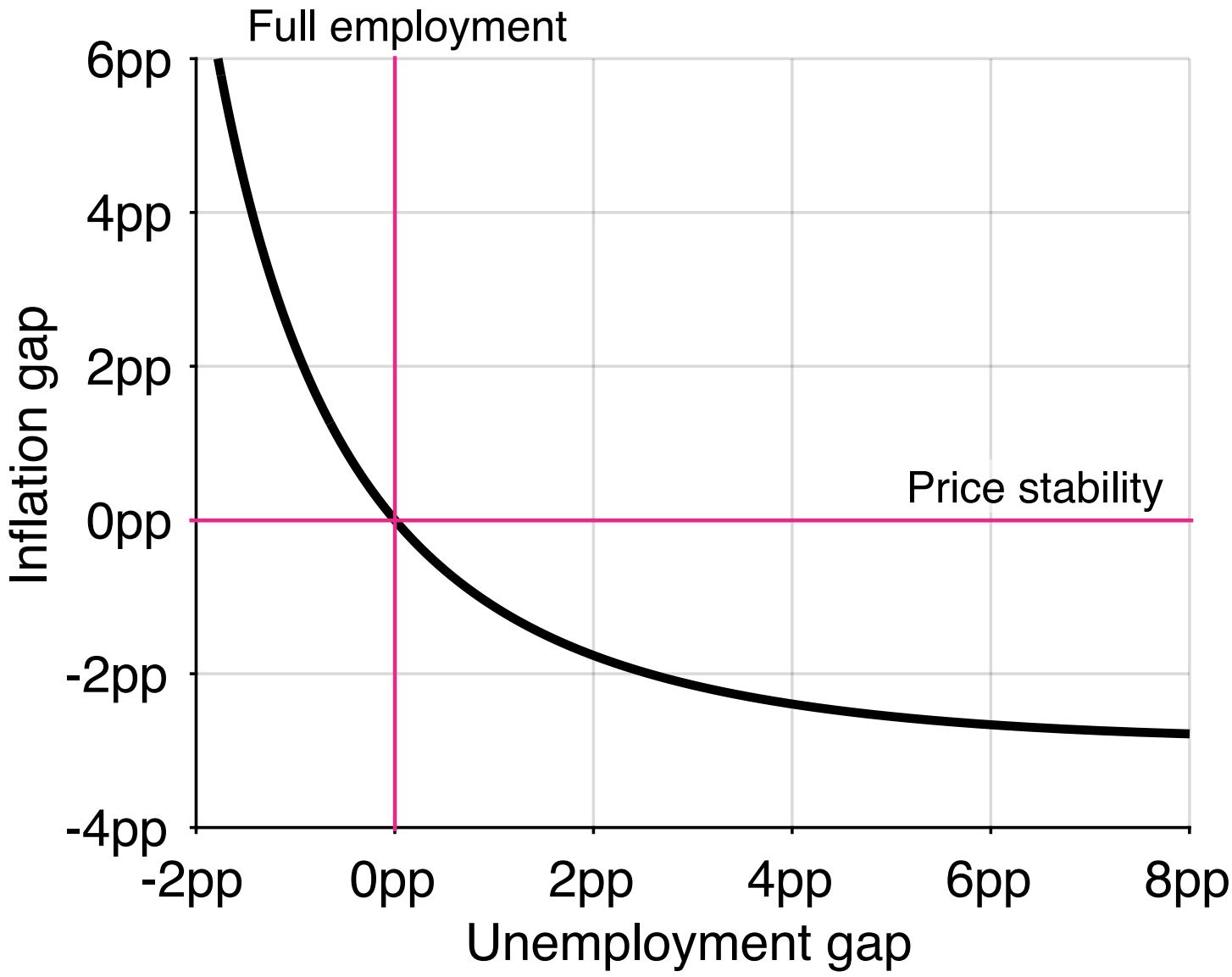
  $\rightsquigarrow$ coincidence likely to hold (Hazell et al 2022)

# #2 - CONVEXITY IN UNEMPLOYMENT

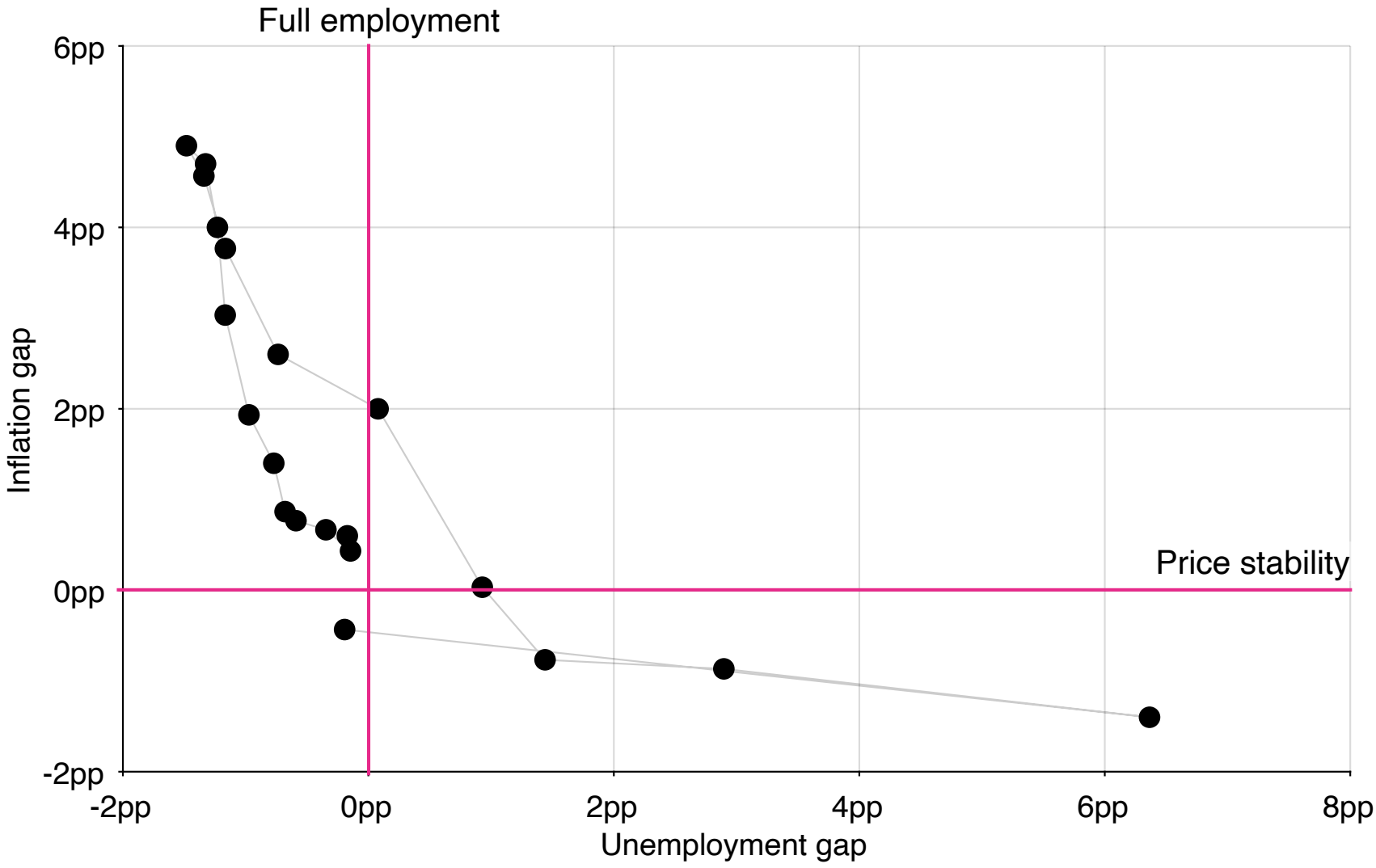
# CALIBRATION: BEVERIDGE CURVE ≈ RECTANGULAR HYPERBOLA



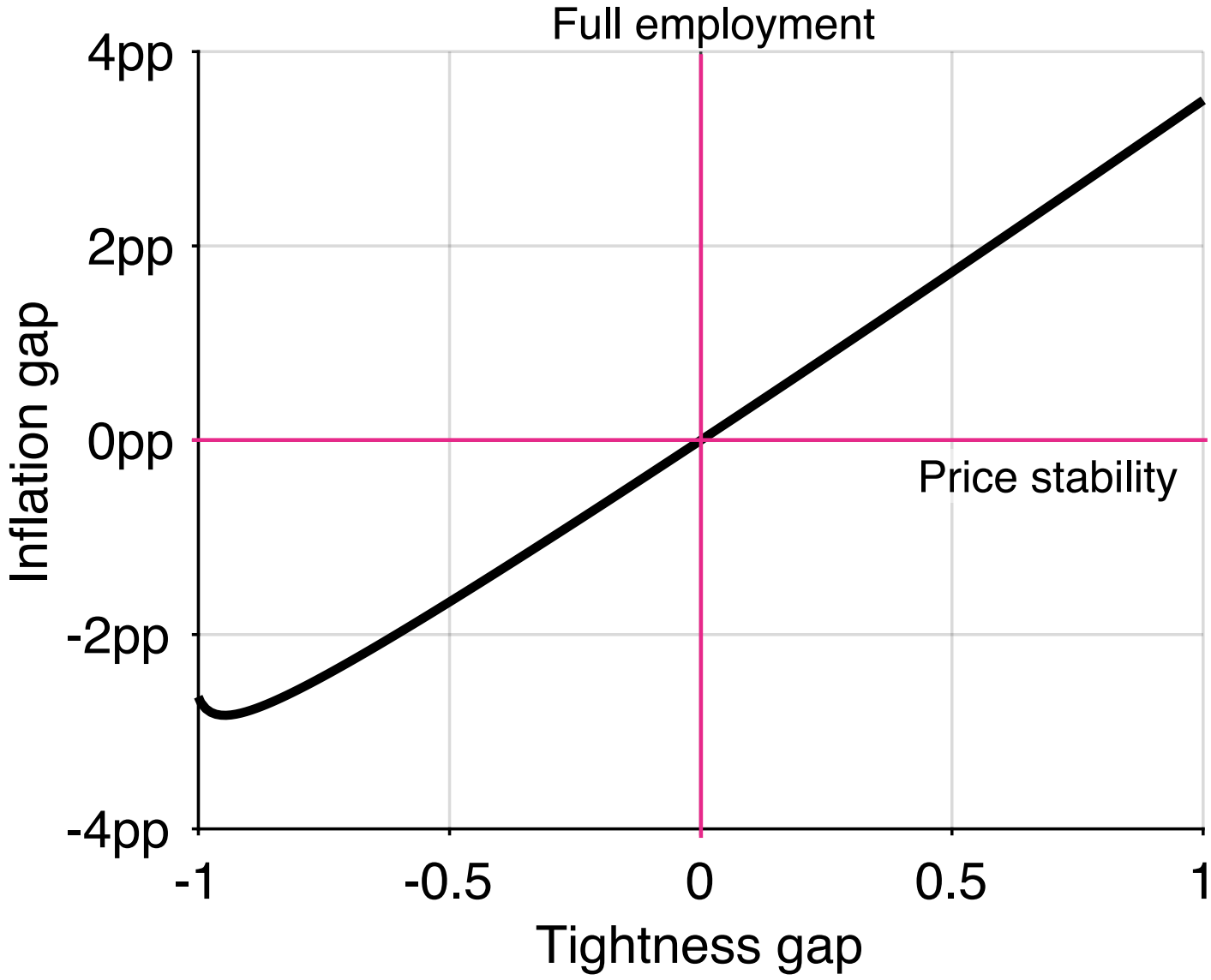
# UNEMPLOYMENT-INFLATION PHILLIPS CURVE IS CONVEX



# SIMILARITY TO POST-PANDEMIC PHILLIPS CURVE

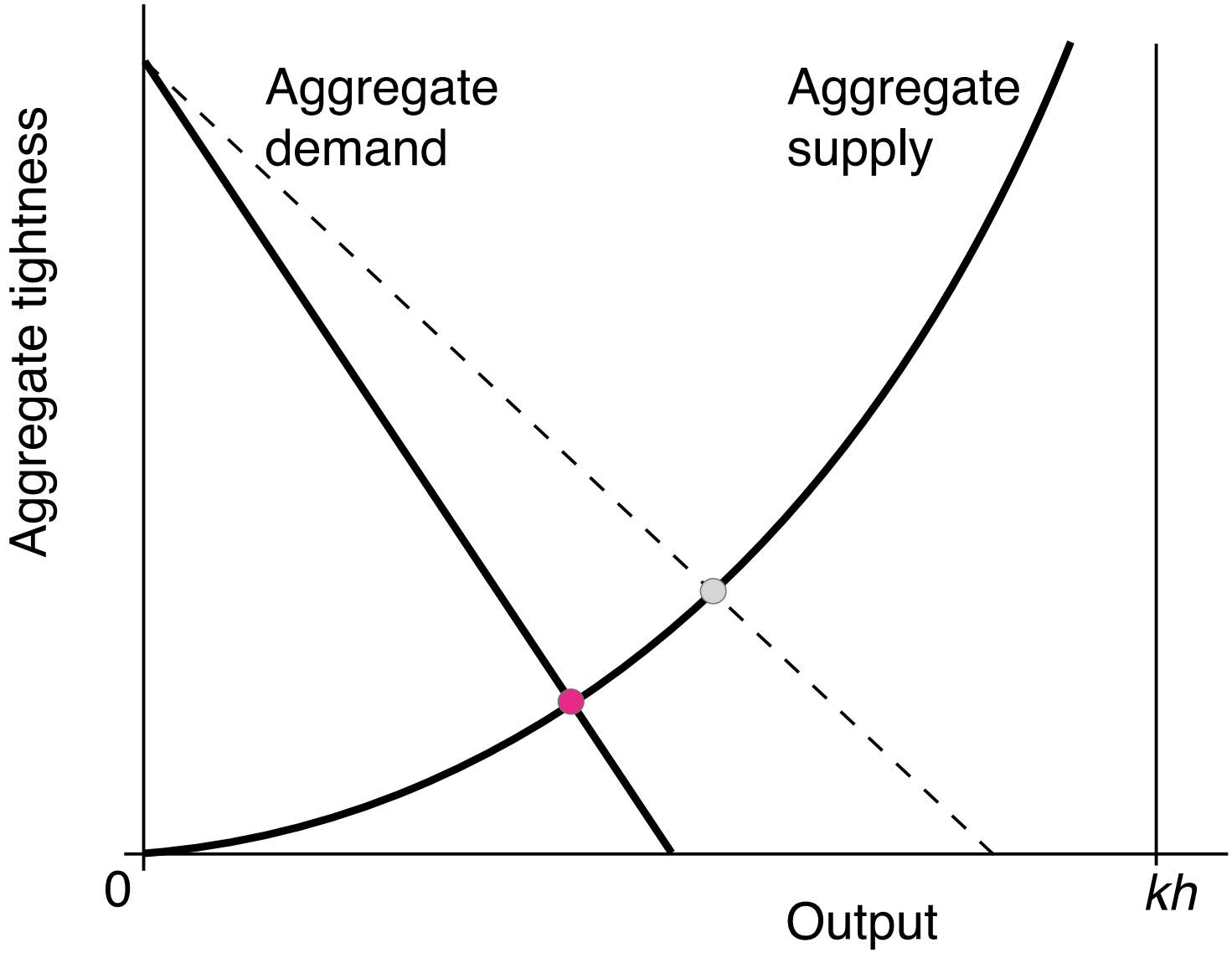


# BUT TIGHTNESS-INFLATION PHILLIPS CURVE IS NEAR LINEAR

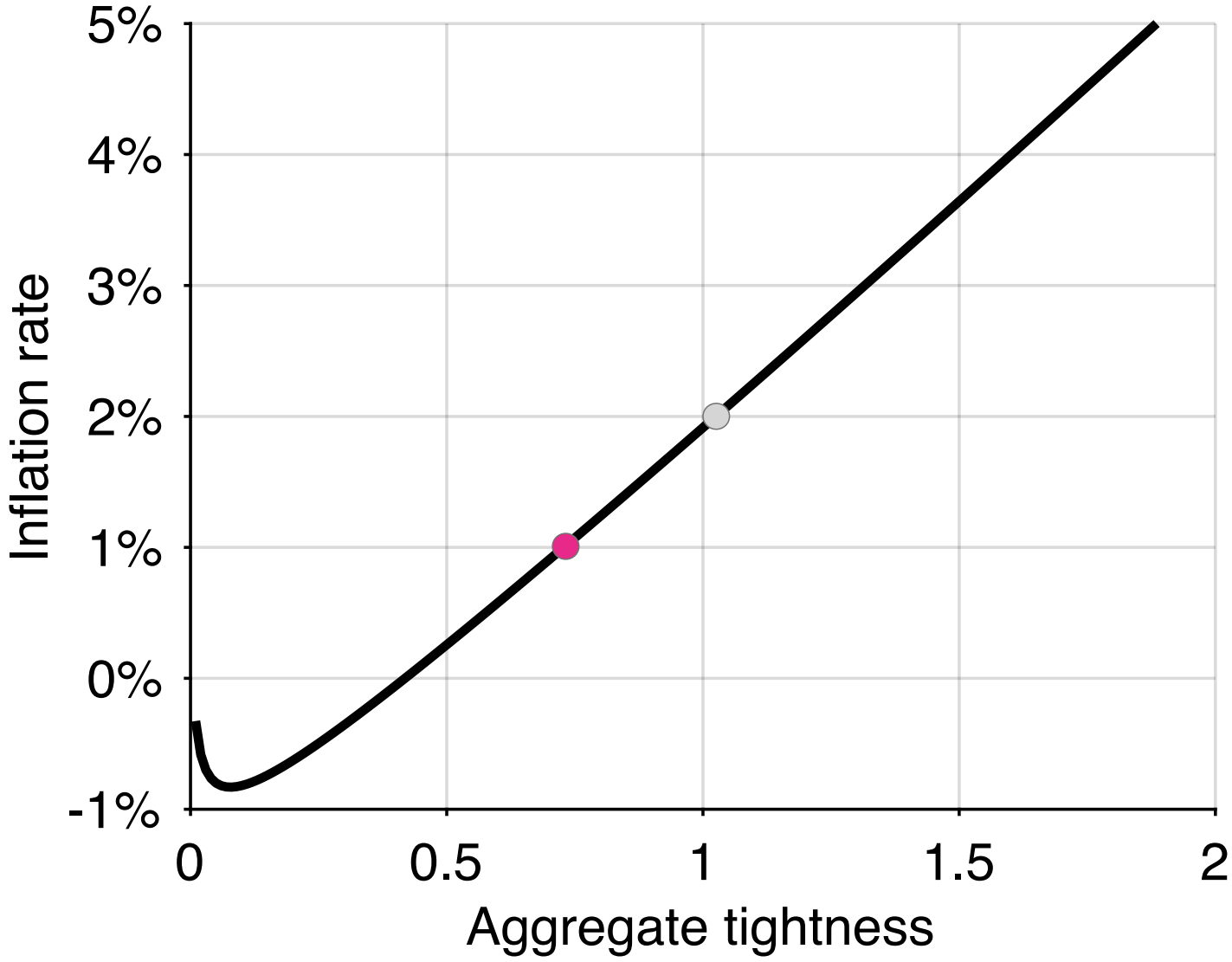


# #3 - SLACK DETERMINES INFLATION

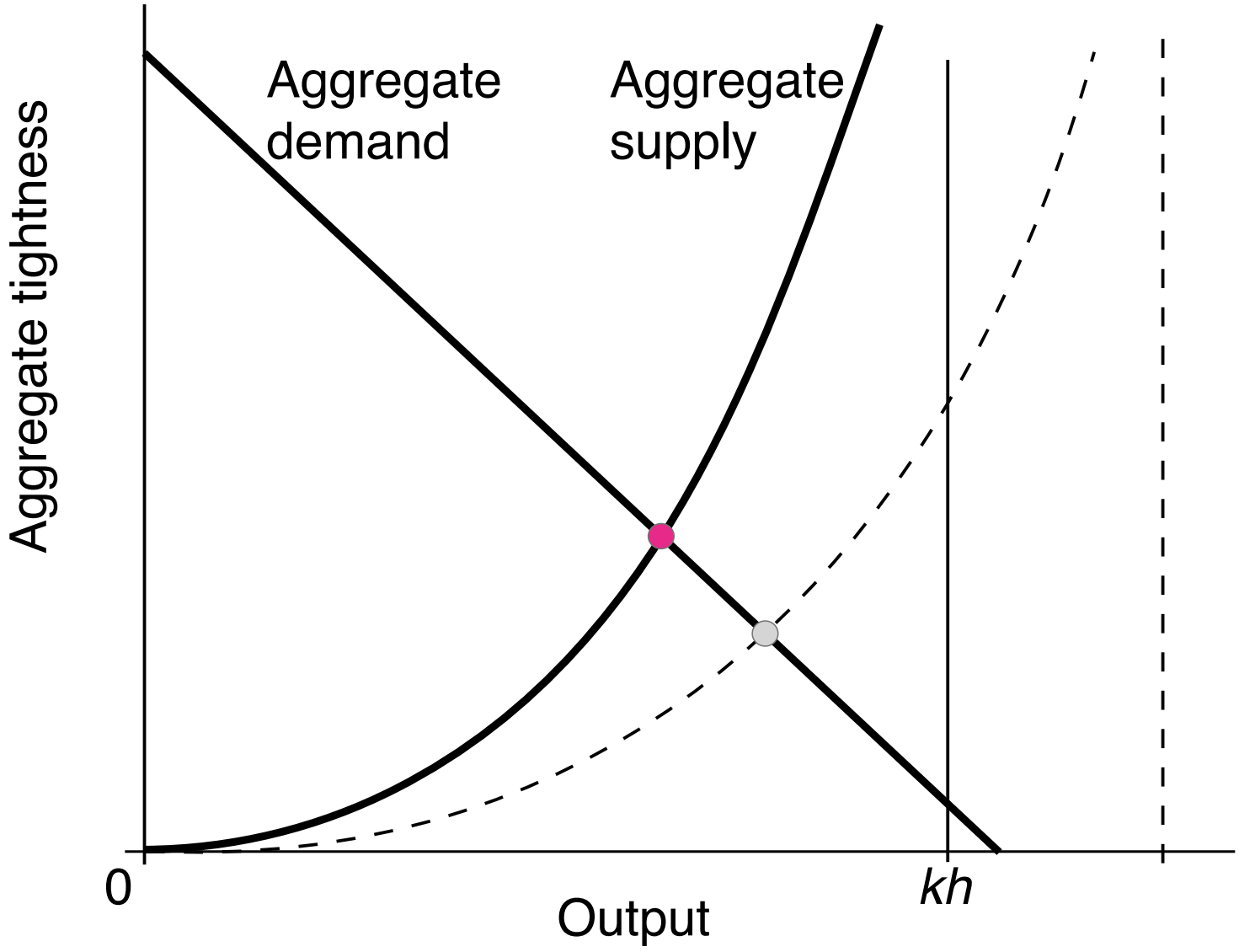
# RECESSION: NEGATIVE AD SHOCKS ⇝ LOWER TIGHTNESS



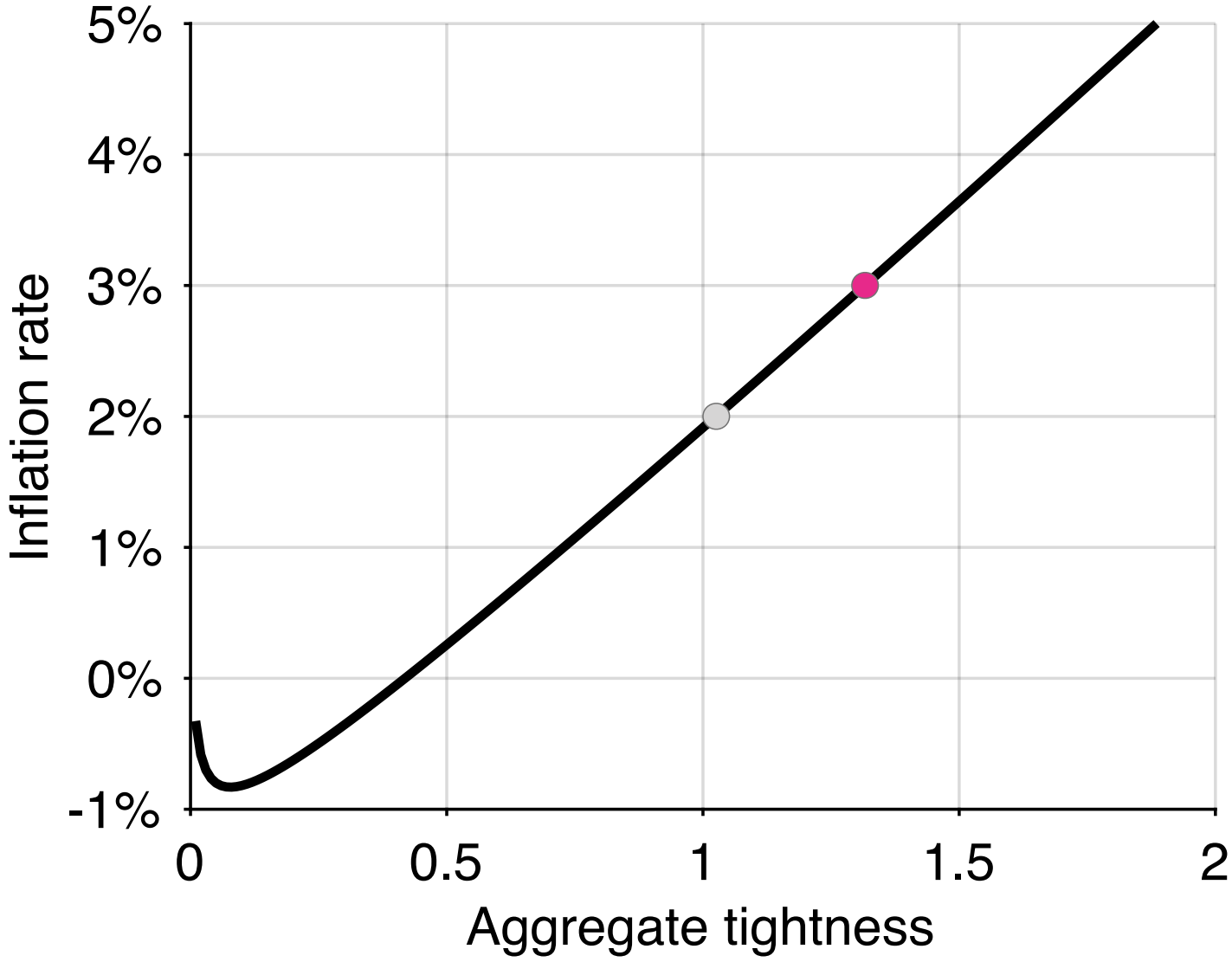
# RECESSION: NEGATIVE AD SHOCKS ⇝ LOWER INFLATION



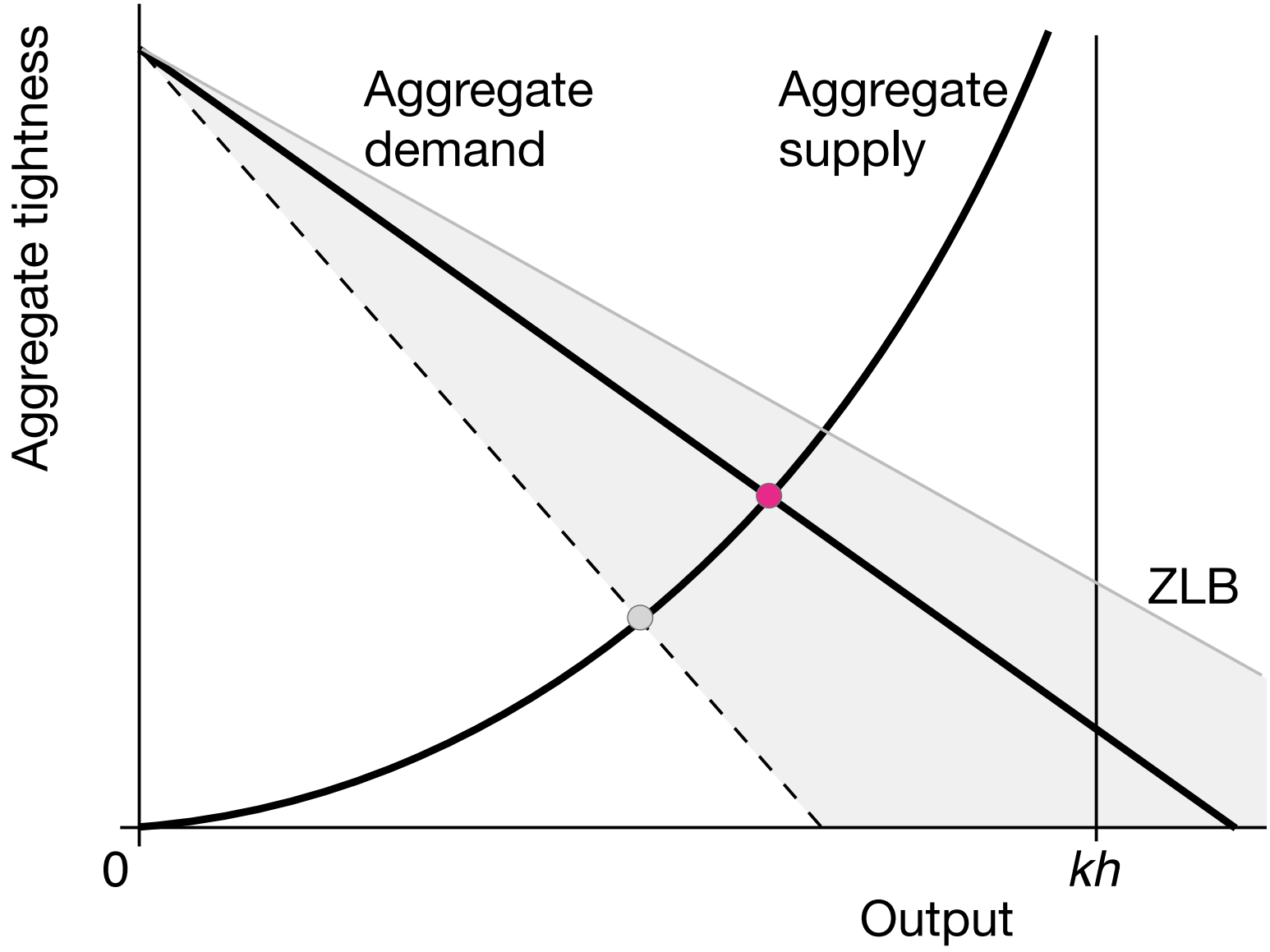
# STAGFLATION: NEGATIVE AS SHOCKS ⇝ HIGHER TIGHTNESS



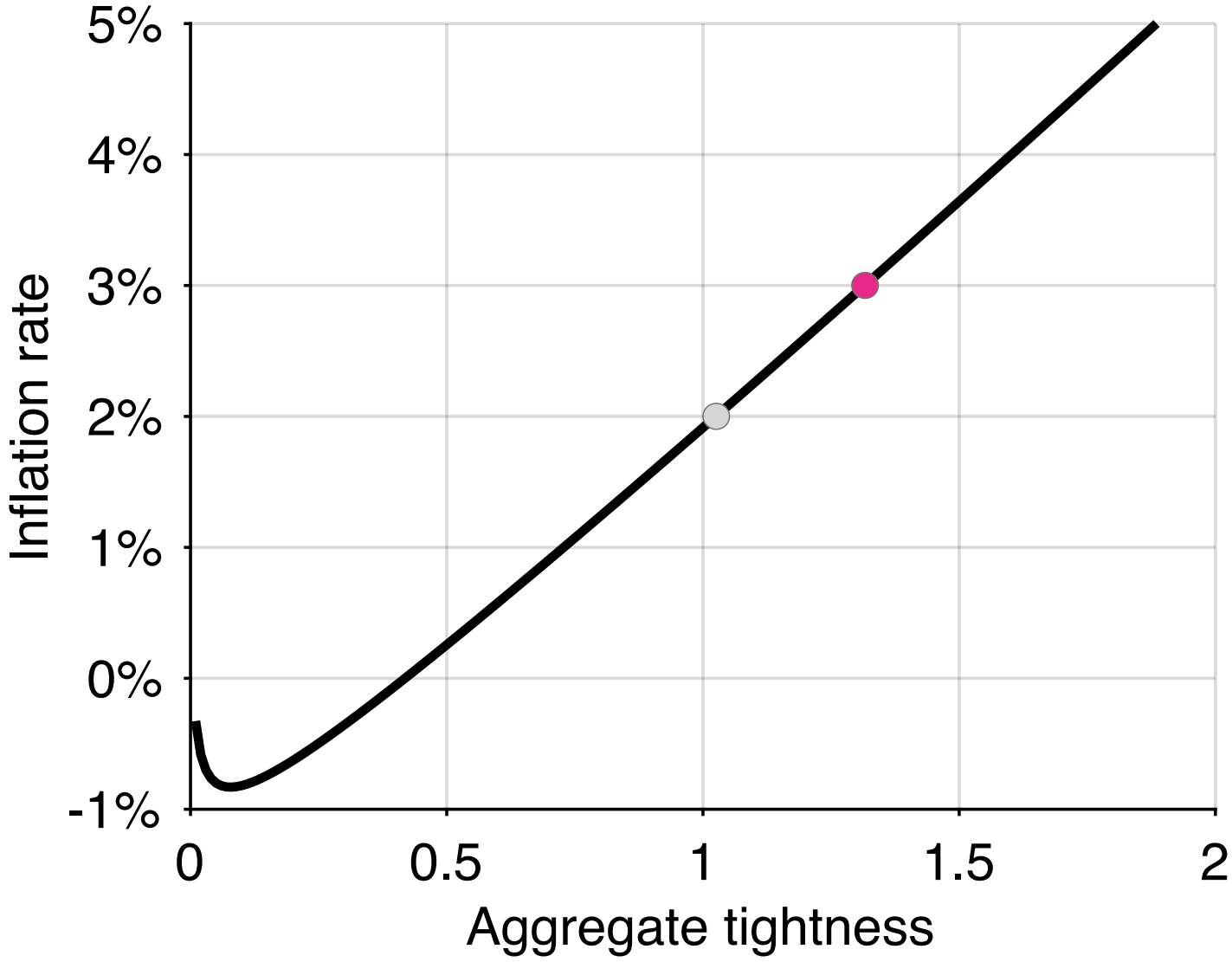
# STAGFLATION: NEGATIVE AS SHOCKS ⇝ HIGHER INFLATION



# POLICY: INTEREST-RATE CUTS ⇝ HIGHER TIGHTNESS

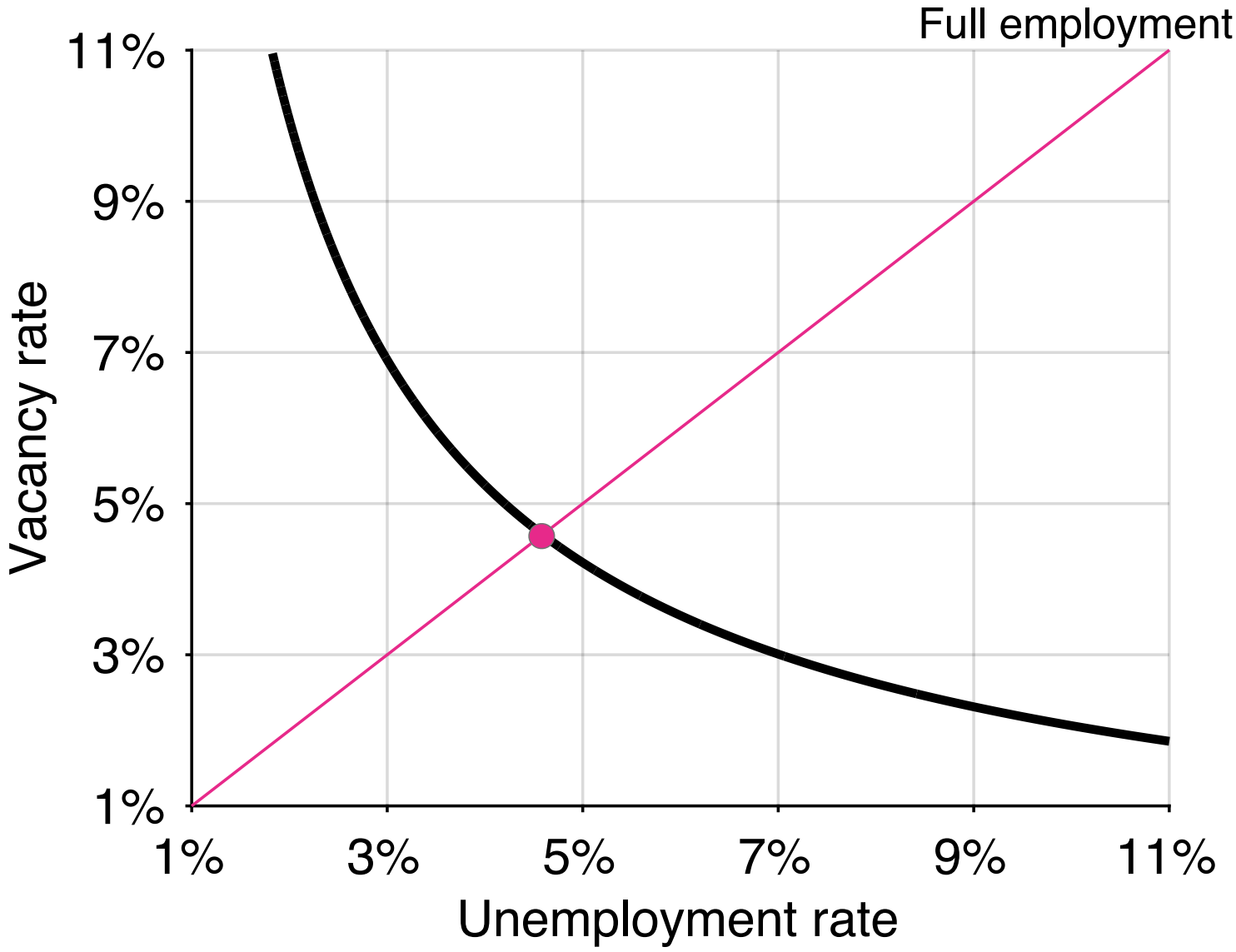


# POLICY: INTEREST-RATE CUTS ⇝ HIGHER INFLATION

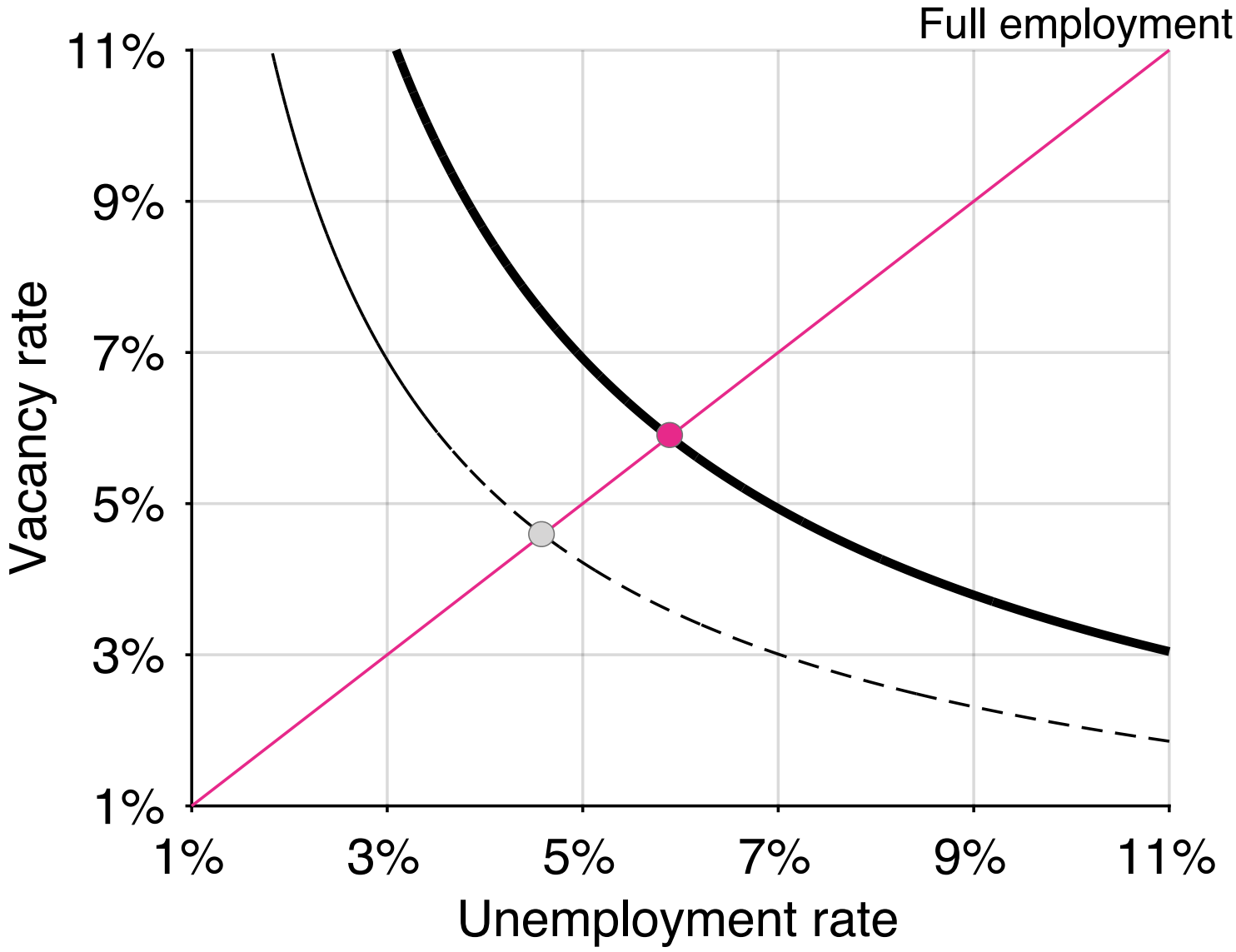


# #4 - BEVERIDGE SHOCKS ARE “COST-PUSH” SHOCKS

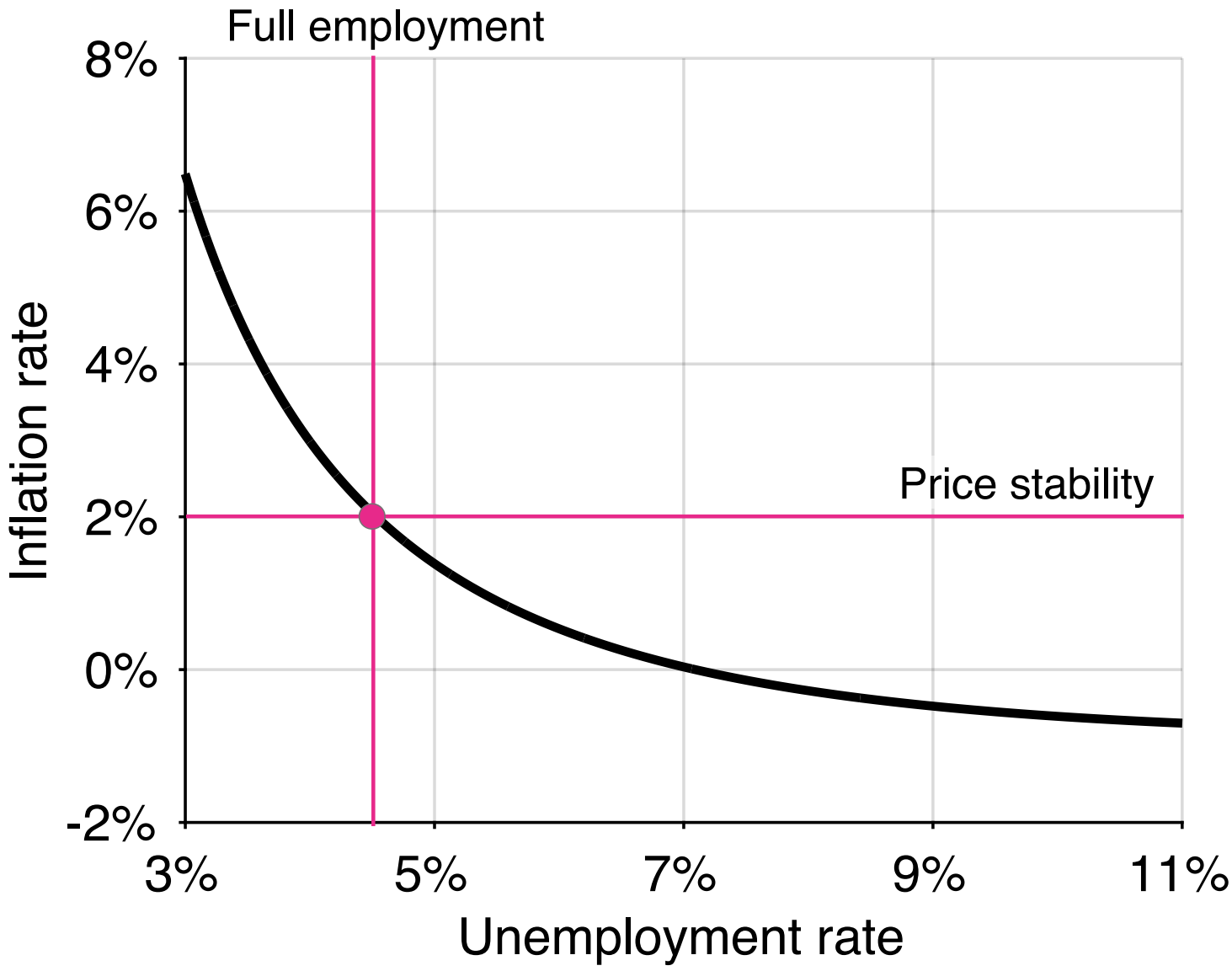
# SHIFTS IN BEVERIDGE CURVE RAISE FERU . . .



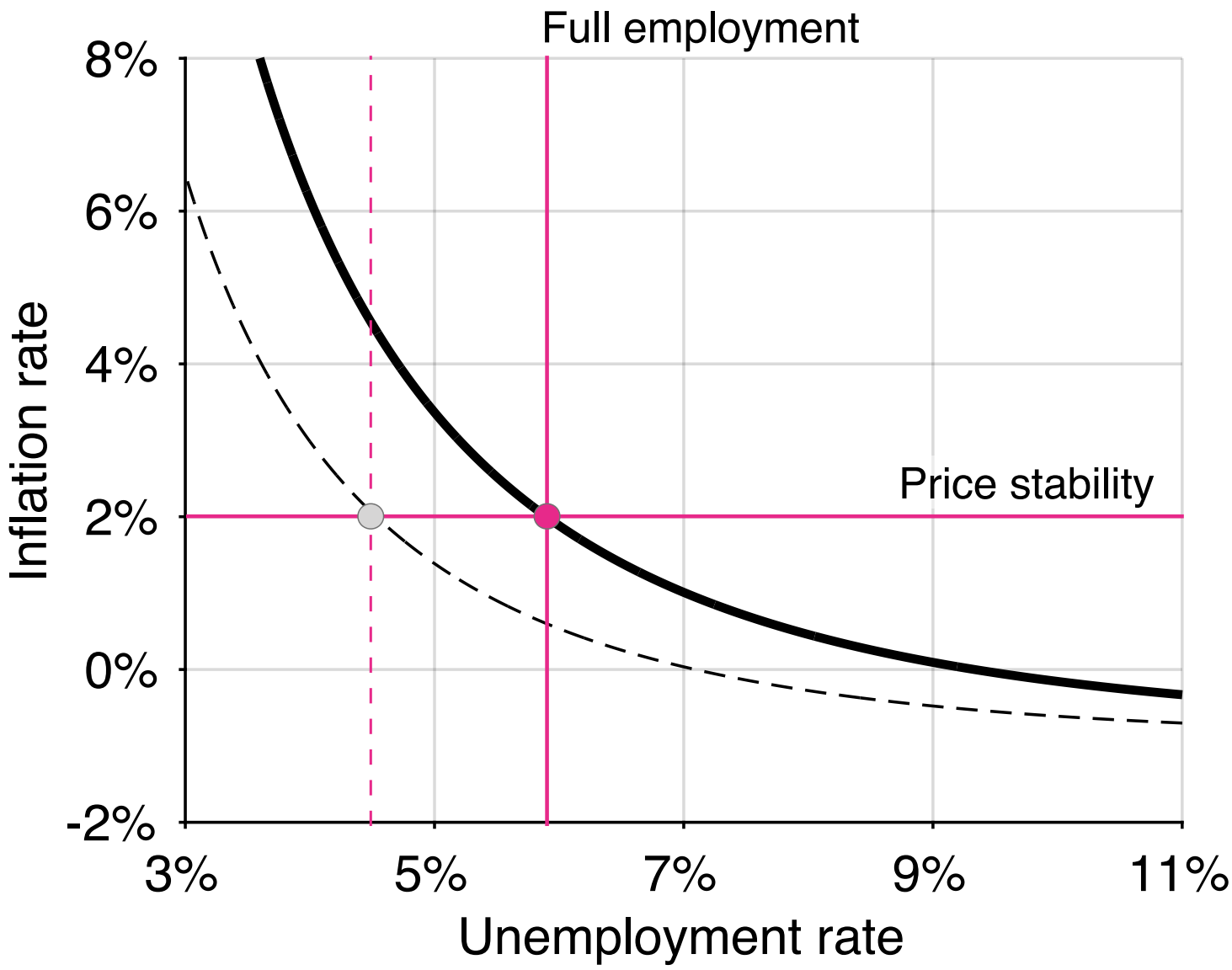
# SHIFTS IN BEVERIDGE CURVE RAISE FERU . . .



# . . . WHICH SHIFTS PHILLIPS CURVE

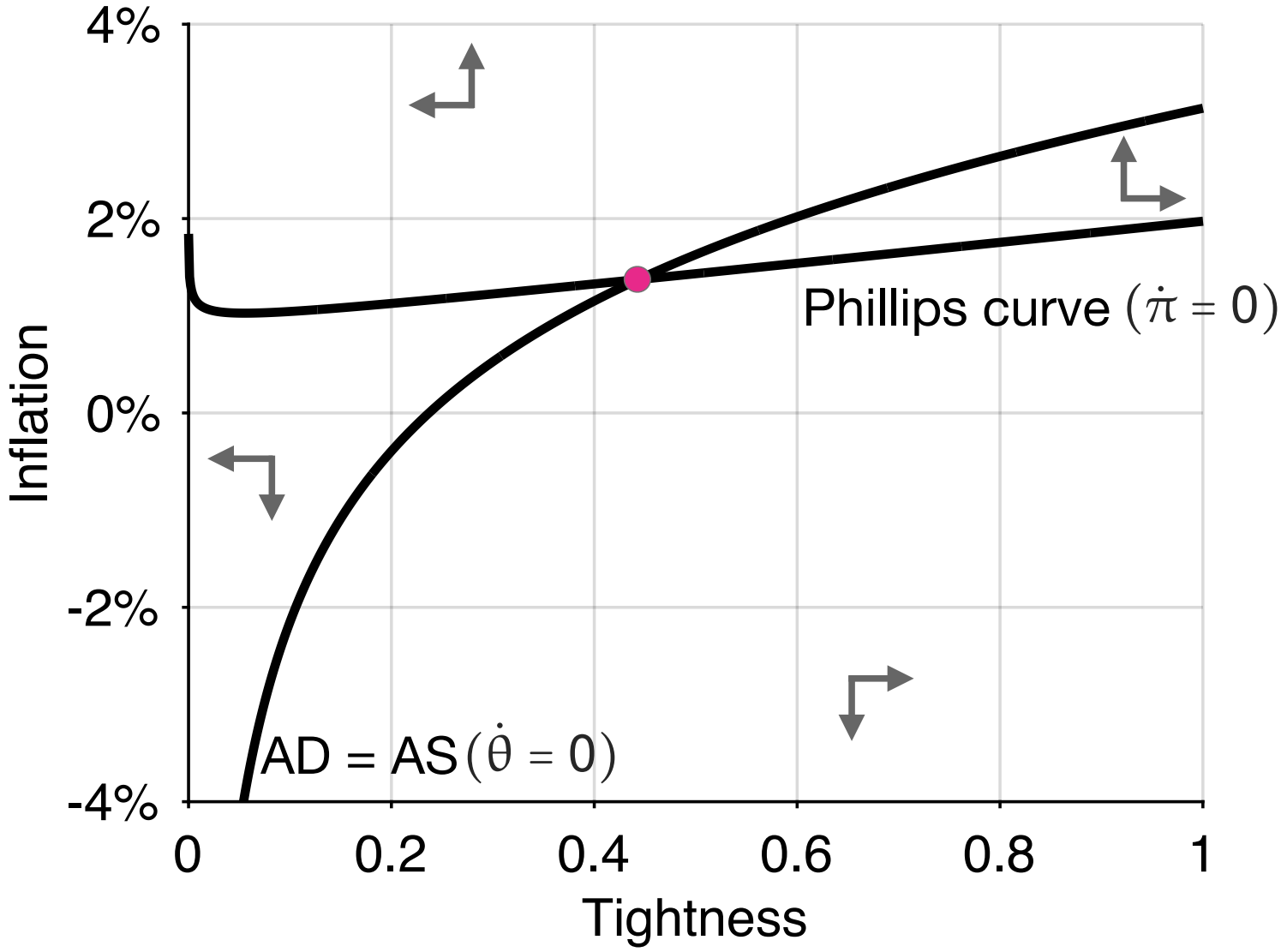


# . . . WHICH SHIFTS PHILLIPS CURVE



# #5 - NORMAL, UNIQUE SOLUTION AT ZLB

# PHASE DIAGRAM AT THE ZLB



# NEGATIVE DEMAND SHOCK AT THE ZLB

